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DT081A – Signal and Image Processing

Prof Mårten Sjöström

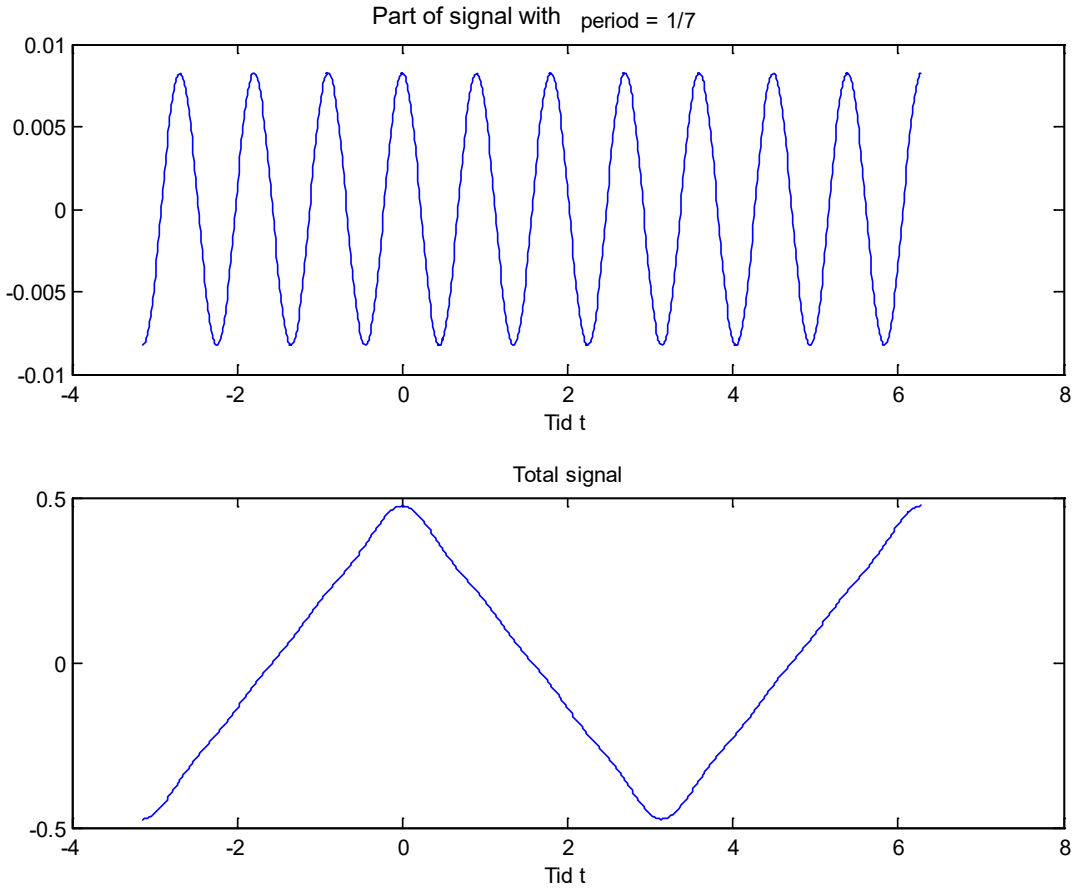


Lecture 4

Time-Domain Analysis



Sum of Sinusoids



Creates a triangle wave!

Exampels (webb)

Sum of Sinusoids

- The principle of superposition:
  - The sum of many sinusoids forms another periodic signal

- Advantage
  - It is easier to handle (calculate, understand) a sum of sinusoids than a general periodic signal



Fourier Series

- The Fourier Series is a sum of complex sinusoids
  - $x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}, \quad C_k \in \mathbb{C}$ (inverse Fourier series – periodic functions)

    $e^{jx} = \cos(x) + j\sin(x)$
- The Fourier coefficients are given by
  - $C_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-jk\omega_0 t}dt,$ $T_0 = \frac{2\pi}{\omega_0}$ = period of function



Fourier Transform

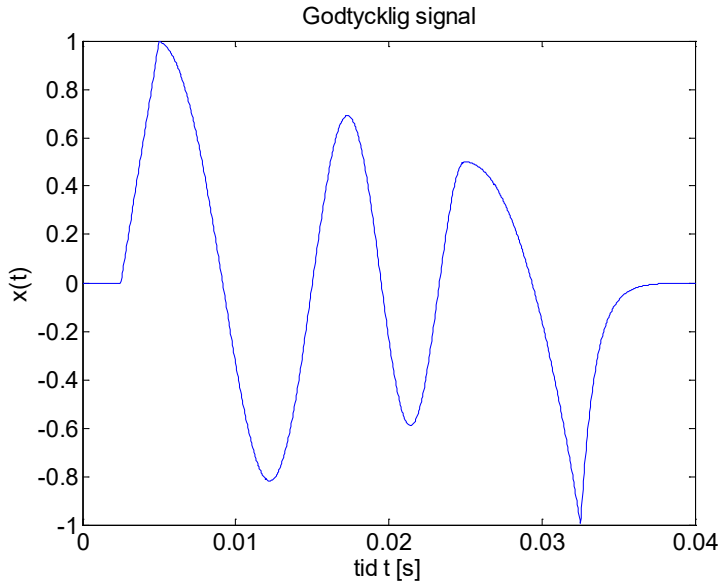
- The inverse Fourier Transform is integral of complex sinusoids
  - $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$ (inverse Fourier Transform – non-periodic functions)
  - Each signal consists of an infinite sum of sinusoids of all kinds of frequencies

- The Fourier Transform is given by
  - $X(\omega) = \langle x(t), e^{j\omega t}\rangle = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \in \mathbb{C}$ (Fourier transform)
    - $X(\omega) = |X(\omega)|e^{j \arg X(\omega)} \in \mathbb{C}$ (Magnitude and Phase)
- The principle of superposition
  - The sum of many sinusoids of all kinds of frequencies forms a general signal.
  - The Fourier transform is of central importance for the understanding of signals and systems.

We make it simple

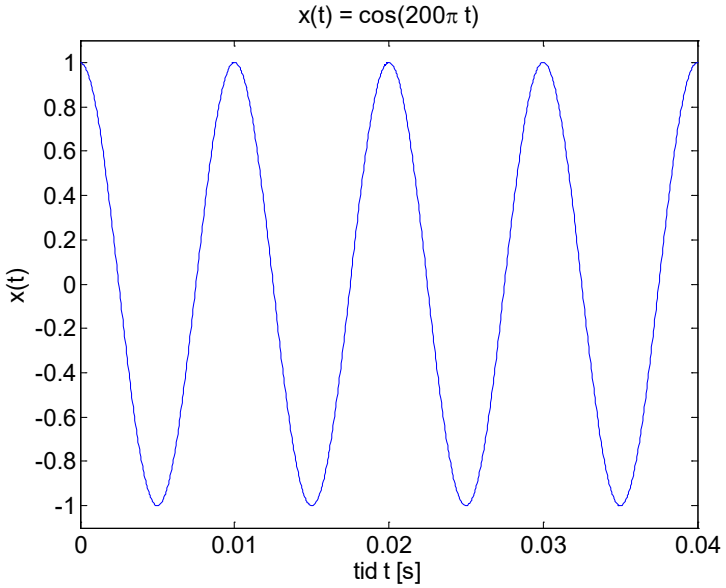
Example 1

- An arbitrary signal
  - Is difficult to compute



Example 2

- A sinusoid
  - Is simpler to compute
  - Fully described by:
    - Amplitude
    - Frequency
    - Phase

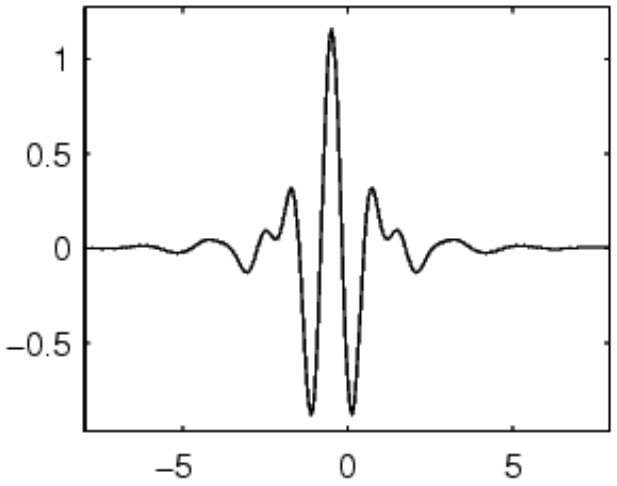


We make it simple

- We make it easy for ourselves by describing signals using functions that are easy to calculate, easy to understand, or that have special properties.
- These are called base functions. (AH!)

Examples

- Sinusoids (Fourier Analysis)
- Wavelets
  - small waves with special properties

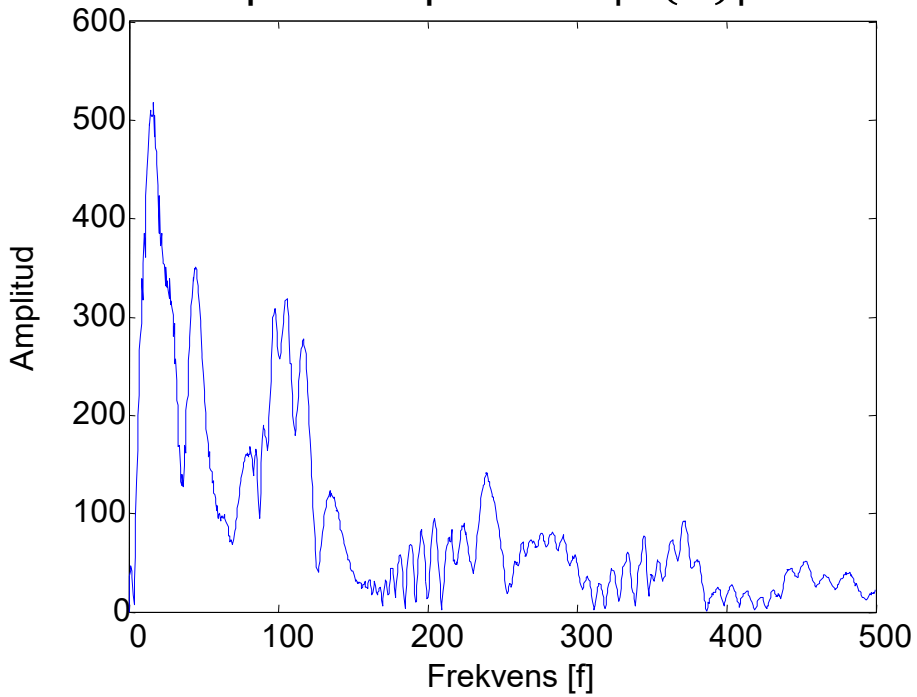


Wavelet function psi

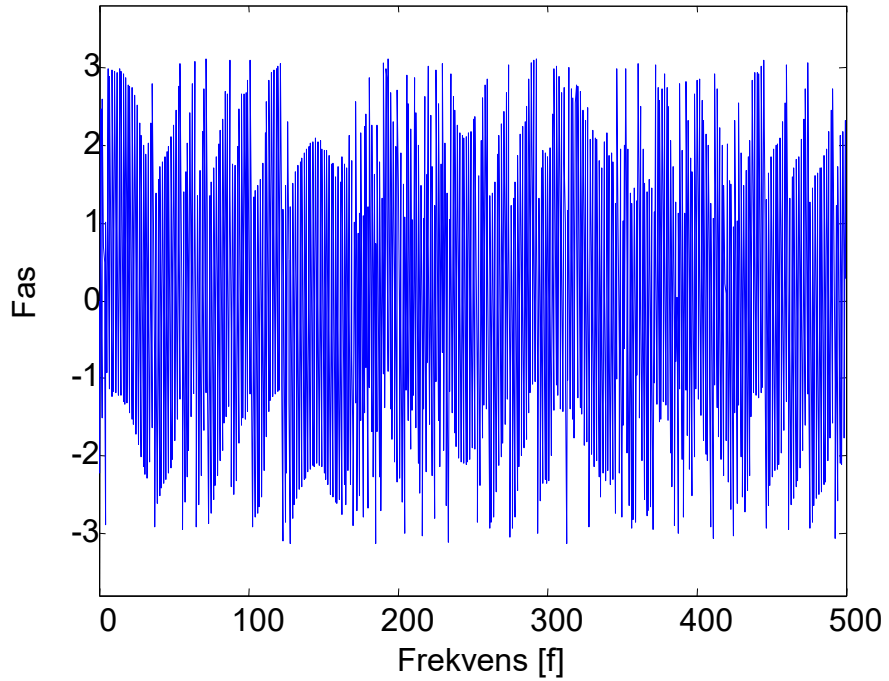
Signal Spectrum

- A signal spectrum describes a signal's frequency content: which sinusoids it consists of
  - Amplitude spectrum $|X(\omega)| = \sqrt{X(\omega)X^*(\omega)}$
  - Phase spectrum $\arg X(\omega) = \operatorname{atan}\frac{Im\,X(\omega)}{Re\,X(\omega)}$ $(+\pi$ if $Re\,X(\omega)\,)$

Amplitude spectrum $|X(\omega)|$



Phase spectrum $\arg X(\omega)$



Fourier Transform – Negative frequencies?

- A signal consists of integrated sinousois (Inverse Fourier Transform)
  - $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}dt, \quad X(\omega) \in \mathbb{C}$

- $-\infty < \omega < \infty \Rightarrow$ Negative frequencies !
  - Do they exist?

- No, they are only a consequence of a mathematical formulation
  - The energy for a certain frequency is divided between $+\omega_1$ and $-\omega_1$

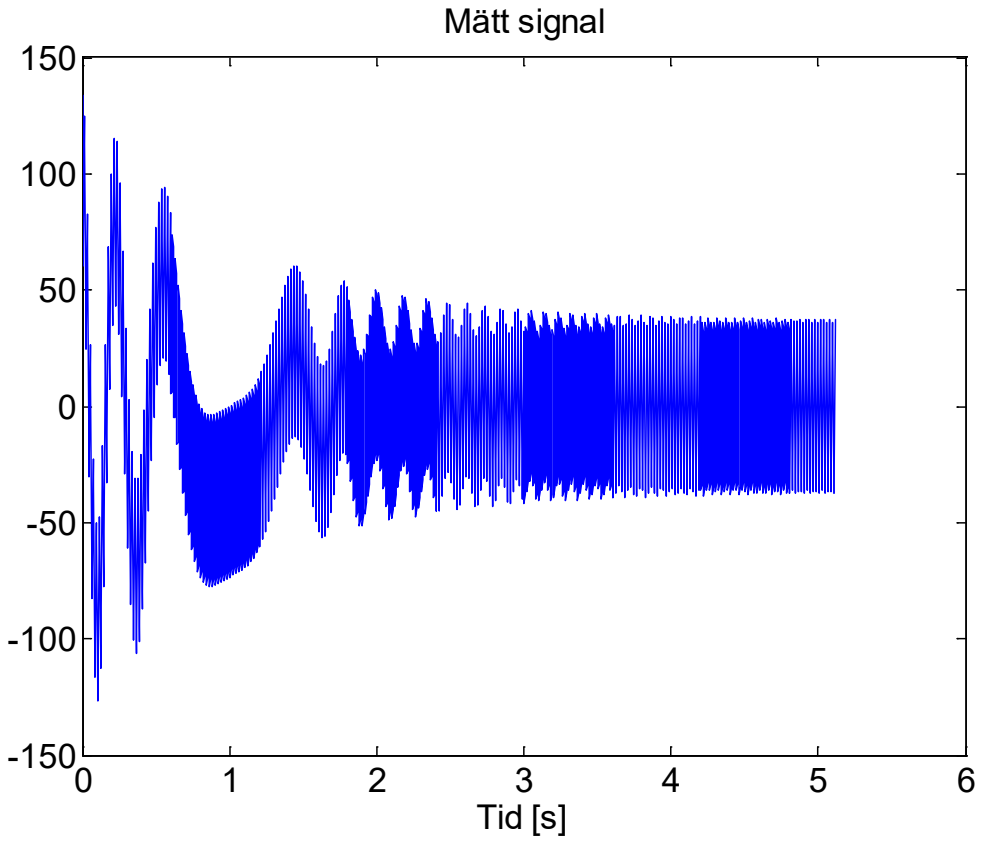


The Principle of Filtering

- In filtering, interesting information is extracted by selecting interesting frequencies.

Example

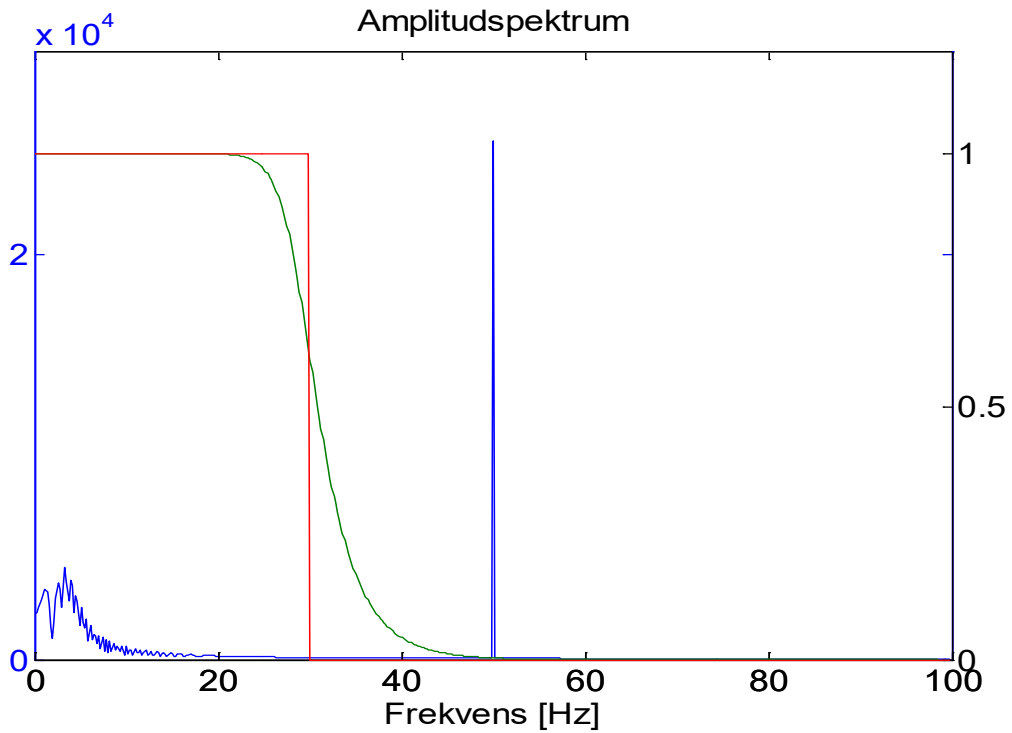
- Measured signal with 50 Hz "hum" from the power source.



The Principle of Filtering

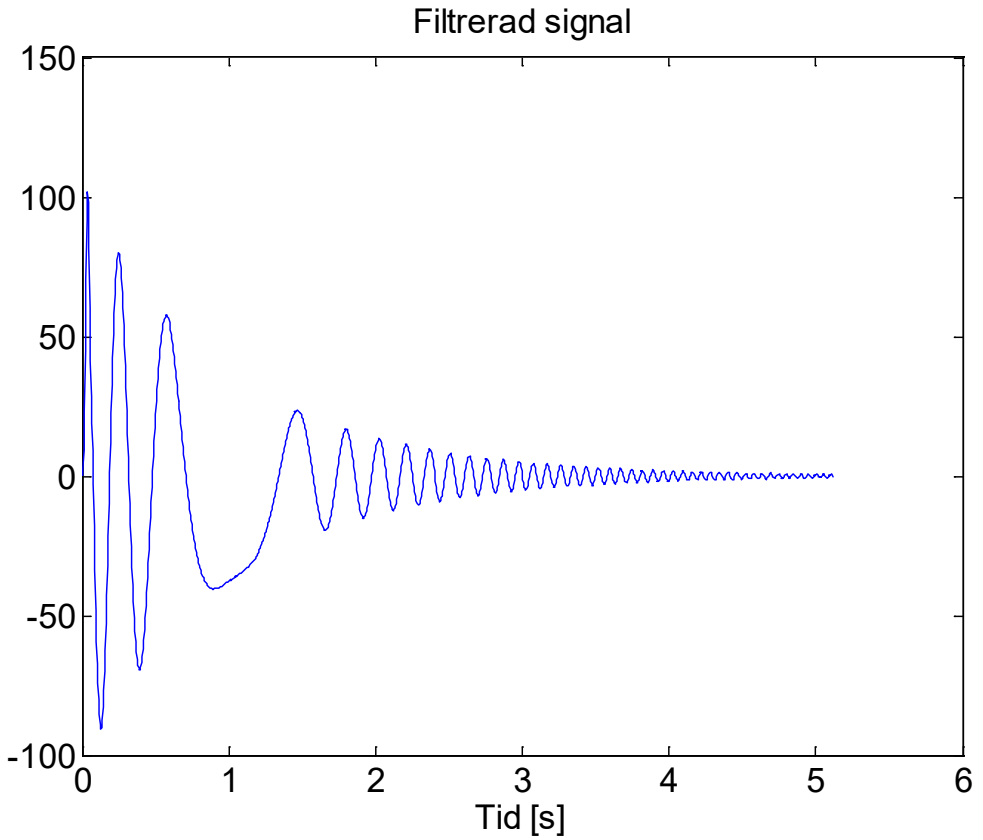
- The spectrum of the signal
  - The frequency content of the signal

- Low pass filter
  - Red: ideal filter
  - Green: real filter



The Principle of Filtering

- Filtered signal

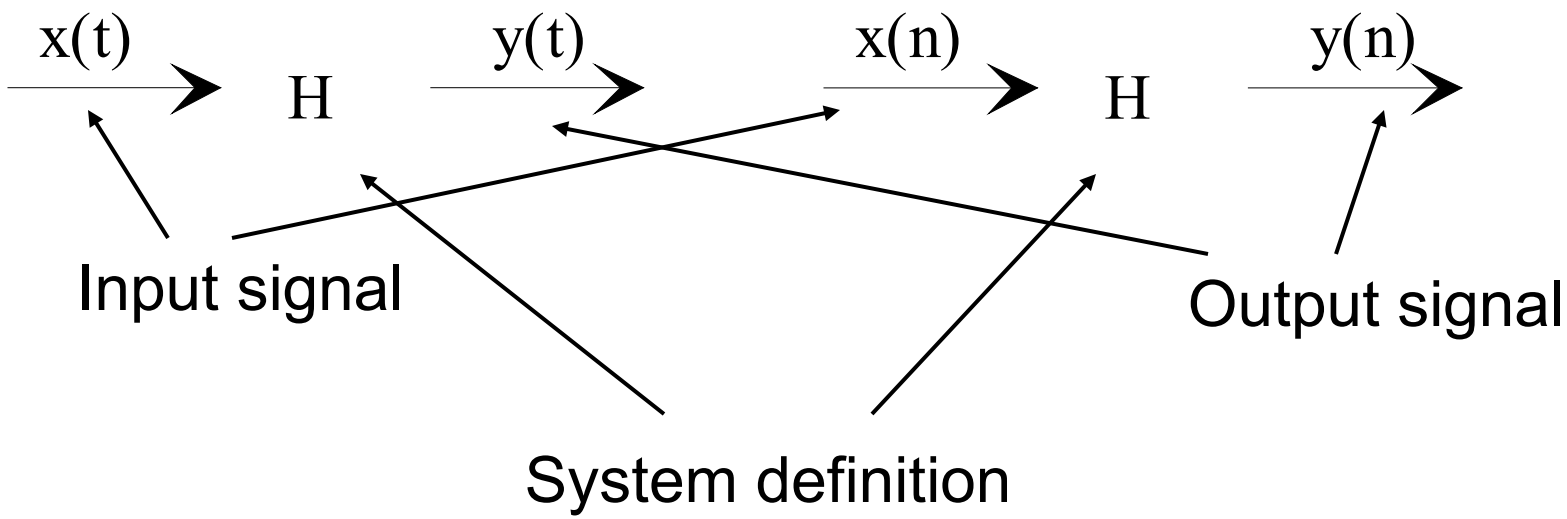


Linear Time-Invariant Systems



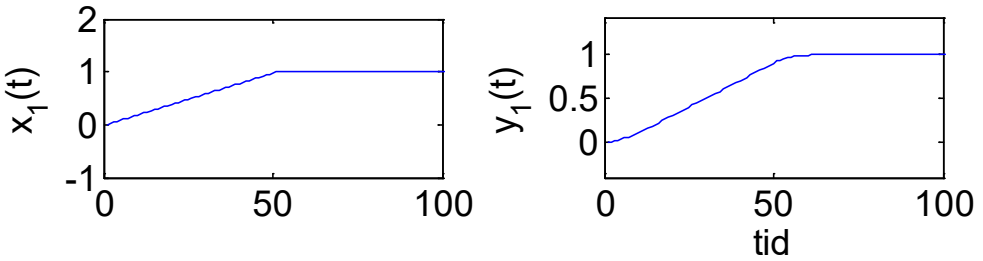
System

- Continuous Time System
- Discrete Time System

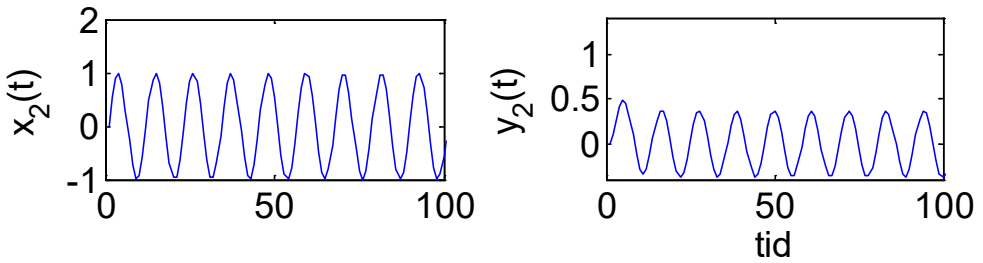


Linear System

$$x = x_1 \rightarrow y = y_1$$



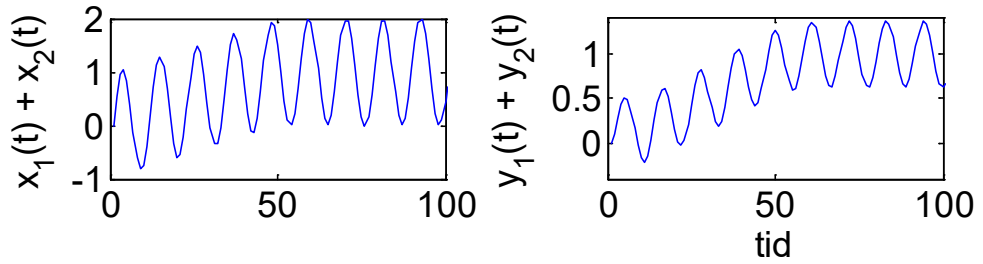
$$x = x_2 \rightarrow y = y_2$$



$$x = a \cdot x_1 + b \cdot x_2$$

$$\rightarrow$$

$$y = a \cdot y_1 + b \cdot y_2$$



Causal System

- A system,
  - where cause comes before effect, is causal,
  - otherwise it is non-causal

- All physical systems are causal!
- Models can be non-causal.
- Example
  - Average value

$$y(n) = \frac{1}{3}\left[x(n+1) + x(n) + x(n-1)\right]$$

  - *This is a non-causal system – it depends on values in the future!*



Time-Invariant / Shift-Invariant System

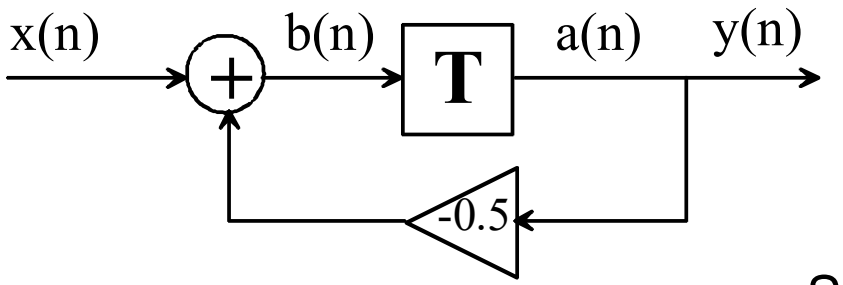
- A system that does not depend on the time at which the input signal was applied,
  - i.e. if $x(t)$ gives the output signal $y(t)$
  - then $x(t - t_0)$ gives the output signal $y(t - t_0)$

- ”Our” systems are
  - Linear and Time-Invariant – LTI Systems $\Rightarrow$ Linear Transforms can be applied
  - Often they are also *Causal*



Example – Discrete Time System

- Discrete Time System – Derivation



Auxiliary variable (add after sum)

$$b(n) = a(n+1) = x(n) - 0.5\,y(n)$$

$$y(n) = a(n)$$

State variable (add after time delay)

Linear difference equation

Output of an LTI System

- The output from a system is given by a convolution
  - $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$ (Continuous Time System)
  - $y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$ (Discrete Time System)

- An LTI system is fully defined by
  - Its *impulse response* $h(t)$ (time continuous) $h(k)$ (time discrete)
  - [Or its *transfer function* $H(s) = L\{h(t)\}$ (Laplace Transform) $H(z) = Z\{h(k)\}$ (Z-Transform)]



Output of an LTI System by an Impulse Input

- Let continuous time system input be an impulse
  - $x(t) = \delta(t)$ (impulse)
    $= \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$ where $\int_{-\infty}^{\infty} \delta(t)dt = 1,$
    - For which: $\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$

- The system output is then
  - $y(t) = h(t) * \delta(t)$
    $= \int_{-\infty}^{\infty} h(\tau) \cdot \delta(t - \tau)d\tau$
    $h(t)$

    (impulse response)

- Let discrete time system input be a unit pulse
  - $x(n) = \delta_{\mathrm{K}}(n)$ (unite pulse)
    $= \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$

- The system output is then
  - $y(n) = h(n) * \delta_{\mathrm{K}}(n)$
    $= \sum_{k=-\infty}^{\infty} h(k) \cdot \delta_{\mathrm{K}}(n - k) =$
    $h(n)$

    (unit pulse response, a.k.a. impulse resp.)



Consequences for LTI systems – Example

- An LTI system *preserves frequencies*
  - That is, the output contains *only the same frequencies* as in the input signal
  - These frequencies can in theory be infinitely damped (amplitude = 0)

- An LTI system affects only the input signal
  - amplitude and
  - phase
    – for each frequency

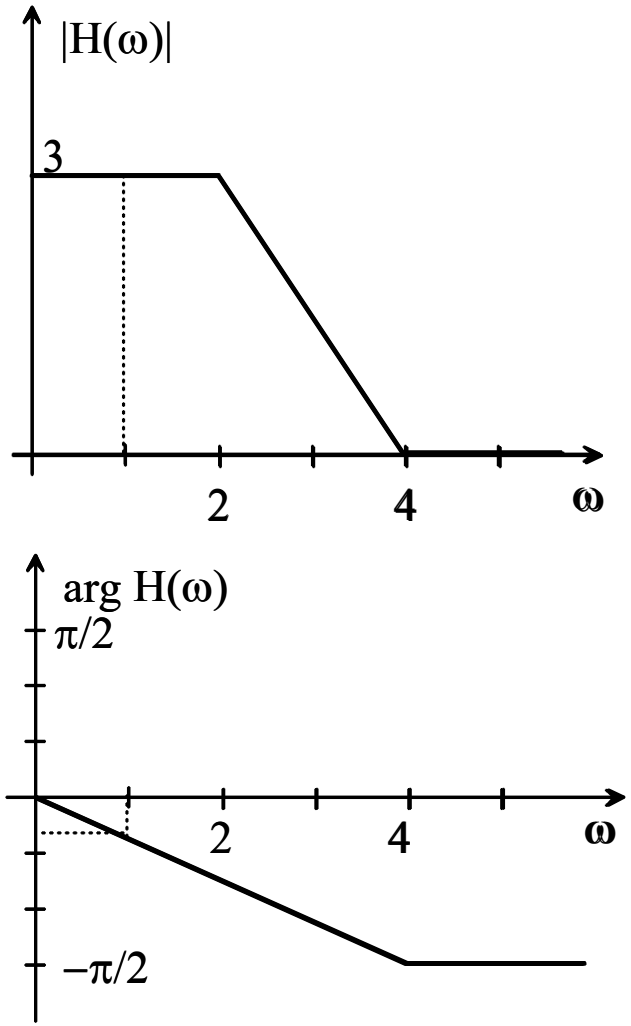


Example LTI system

- Input is $x(t) = 2\sin(1 \cdot t + \pi/4)$
- The LTI system is frequency preserving

$\Rightarrow$

- Amplitude
  - $A_y = |H(1)| \cdot |X(1)|$
    $= 3 \cdot 2 = 6$
- Phase
  - $\phi_y = \arg H(1) + \arg X(1)$
    $= -\pi/8 + \pi/4 = \pi/8$
- Therefore, the output is
  - $y(t) = 6\sin(1 \cdot t + \pi/8)$



Causual System

- A casual discrete time LTI system has an impulse response such that
  - $h(n) = \begin{cases} h^+(n), & n \geq 0 \\ 0, & n < 0 \end{cases}$

$$\Rightarrow \quad y(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k) = \sum_{k=0}^{\infty} h^+(k) \cdot x(n-k)$$



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